

TASK 6: Sales Trend Analysis Using Aggregations

```
17 -- View table
18 • SELECT * FROM sales_transactions;
19 • SELECT count(*) FROM sales_transactions;
20
21 -- Load CSV into table
22 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/Online Sales Data.csv'
23 INTO TABLE sales_transactions
```

transaction_id	transaction_date	product_category	product_name	units_sold	unit_price	total_revenue	region	payment_method
10001	2024-01-01	Electronics	iPhone 14 Pro	2	999.99	1999.98	North America	Credit Card
10002	2024-01-02	Home Appliances	Dyson V11 Vacuum	1	499.99	499.99	Europe	PayPal
10003	2024-01-03	Clothing	Levi's 501 Jeans	3	69.99	209.97	Asia	Debit Card
10004	2024-01-04	Books	The Da Vinci Code	4	15.99	63.96	North America	Credit Card
10005	2024-01-05	Beauty Products	Neutrogena Skincare Set	1	89.99	89.99	Europe	PayPal
10006	2024-01-06	Sports	Wilson Evolution Basketball	5	29.99	149.95	Asia	Credit Card
10007	2024-01-07	Electronics	MacBook Pro 16-inch	1	2499.99	2499.99	North America	Credit Card
10008	2024-01-08	Home Appliances	Blueair Classic 480i	2	599.99	1199.98	Europe	PayPal
10009	2024-01-09	Clothing	Nike Air Force 1	6	89.99	539.94	Asia	Debit Card
10010	2024-01-10	Books	Dune by Frank Herbert	2	25.99	51.98	North America	Credit Card
10011	2024-01-11	Beauty Products	Chanel No. 5 Perfume	1	129.99	129.99	Europe	PayPal
10012	2024-01-12	Sports	Babolat Pure Drive Tennis ...	3	199.99	599.97	Asia	Credit Card
10013	2024-01-13	Electronics	Samsung Galaxy Tab S8	2	749.99	1499.98	North America	Credit Card

```
18
19 • SELECT count(*) FROM sales_transactions;
20 • SELECT * FROM sales_transactions;
21
22
23 -- Load CSV into table
```

count(*)
240

```

30  -- extract month name from date col and add the col to table
31  • ALTER TABLE sales_transactions ADD COLUMN month VARCHAR(20);
32
33  • UPDATE sales_transactions SET month = MONTHNAME(transaction_date);
34
35  • SELECT * FROM sales_transactions;

```

Result Grid Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	transaction_id	transaction_date	product_category	product_name	units_sold	unit_price	total_revenue	region	payment_method	month
▶	10001	2024-01-01	Electronics	iPhone 14 Pro	2	999.99	1999.98	North America	Credit Card	January
	10002	2024-01-02	Home Appliances	Dyson V11 Vacuum	1	499.99	499.99	Europe	PayPal	January
	10003	2024-01-03	Clothing	Levi's 501 Jeans	3	69.99	209.97	Asia	Debit Card	January
	10004	2024-01-04	Books	The Da Vinci Code	4	15.99	63.96	North America	Credit Card	January
	10005	2024-01-05	Beauty Products	Neutrogena Skincare Set	1	89.99	89.99	Europe	PayPal	January
	10006	2024-01-06	Sports	Wilson Evolution Basketball	5	29.99	149.95	Asia	Credit Card	January
	10007	2024-01-07	Electronics	MacBook Pro 16-inch	1	2499.99	2499.99	North America	Credit Card	January
	10008	2024-01-08	Home Appliances	Blueair Classic 480i	2	599.99	1199.98	Europe	PayPal	January
	10009	2024-01-09	Clothing	Nike Air Force 1	6	89.99	539.94	Asia	Debit Card	January
	10010	2024-01-10	Books	Dune by Frank Herbert	2	25.00	51.00	North America	Credit Card	January

sales_transactions61 x Apply

```

37  -- extract year from date col and add the col to table
38  • ALTER TABLE sales_transactions ADD COLUMN year VARCHAR(20);
39
40  • UPDATE sales_transactions SET year = year(transaction_date);
41  • SELECT * FROM sales_transactions;
42

```

Result Grid Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	transaction_id	transaction_date	product_category	product_name	units_sold	unit_price	total_revenue	region	payment_method	month	year
▶	10001	2024-01-01	Electronics	iPhone 14 Pro	2	999.99	1999.98	North America	Credit Card	January	2024
	10002	2024-01-02	Home Appliances	Dyson V11 Vacuum	1	499.99	499.99	Europe	PayPal	January	2024
	10003	2024-01-03	Clothing	Levi's 501 Jeans	3	69.99	209.97	Asia	Debit Card	January	2024
	10004	2024-01-04	Books	The Da Vinci Code	4	15.99	63.96	North America	Credit Card	January	2024
	10005	2024-01-05	Beauty Products	Neutrogena Skincare Set	1	89.99	89.99	Europe	PayPal	January	2024
	10006	2024-01-06	Sports	Wilson Evolution Basketball	5	29.99	149.95	Asia	Credit Card	January	2024
	10007	2024-01-07	Electronics	MacBook Pro 16-inch	1	2499.99	2499.99	North America	Credit Card	January	2024
	10008	2024-01-08	Home Appliances	Blueair Classic 480i	2	599.99	1199.98	Europe	PayPal	January	2024
	10009	2024-01-09	Clothing	Nike Air Force 1	6	89.99	539.94	Asia	Debit Card	January	2024
	10010	2024-01-10	Books	Dune by Frank Herbert	2	25.00	51.00	North America	Credit Card	January	2024

sales_transactions62 x Apply

```

41      -- group by
42
43      -- by monthly transactions
44 •    SELECT month, COUNT(*) AS total_transactions
45      FROM sales_transactions
46      GROUP BY month;
47

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	month	total_transactions
▶	January	31
	February	29
	March	31
	April	30
	May	31
	June	30
	July	31
	August	27

```

48      -- average revenue by month
49 •    SELECT month, avg(total_revenue) AS average_monthly_revenue
50      FROM sales_transactions
51      GROUP BY month;
52

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	month	average_monthly_revenue
▶	January	469.300645
	February	372.530000
	March	414.491613
	April	415.056333
	May	272.757742
	June	246.151667
	July	219.260645
	August	269.559630

52

53 -- group by product name with total revenue

54 • SELECT month, product_name, sum(total_revenue) AS total_revenue

55 FROM sales_transactions

56 GROUP BY month, product_name;

57

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	month	product_name	total_revenue
▶	January	iPhone 14 Pro	1999.98
	January	Dyson V11 Vacuum	499.99
	January	Levi's 501 Jeans	209.97
	January	The Da Vinci Code	63.96
	January	Neutrogena Skincare Set	89.99
	January	Wilson Evolution Basketball	149.95
	January	MacBook Pro 16-inch	2499.99
	January	Blueair Classic 480i	1199.98
	January	Nike Air Force 1	539.94
	January	Dune by Frank Herbert	51.98

Result 66 ×

59 -- group by product name with total revenue

60 • SELECT product_category, sum(total_revenue) AS total_revenue_product_category

61 FROM sales_transactions

62 GROUP BY product_category;

63

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

product_category	total_revenue_product_category
Electronics	34982.41
Home Appliances	18646.16
Clothing	8128.93
Books	1861.93
Beauty Products	2621.90
Sports	14326.52

```

64  -- group_by region with total revenue
65  • SELECT region, sum(total_revenue) AS total_revenue_region
66  FROM sales_transactions
67  GROUP BY region;
68

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	region	total_revenue_region
▶	North America	36844.34
	Europe	21268.06
	Asia	22455.45

```

70  -- group_by product category by unit sold
71  • SELECT product_category, sum(units_sold) AS total_revenue_region
72  FROM sales_transactions
73  GROUP BY product_category;
74
75

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	product_category	total_revenue_region
▶	Electronics	66
	Home Appliances	59
	Clothing	145
	Books	114
	Beauty Products	46
	Sports	88

```

76 -- unique product_names in sales_transactions;
77 • select distinct product_name from sales_transactions;
78 • select count(distinct product_name) as count_product_name from sales_transactions;
79
80

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	count_product_name
▶	232

```

81 -- total revenue by month
82 • SELECT month, SUM(total_revenue) AS total_monthly_revenue
83 FROM sales_transactions
84 GROUP BY month
85 ORDER BY FIELD(month, 'January', 'February', 'March', 'April', 'May', 'June', 'July', 'August', 'September', 'October', 'November', 'December');
86

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	month	total_monthly_revenue
▶	January	14548.32
	February	10803.37
	March	12849.24
	April	12451.69
	May	8455.49
	June	7384.55
	July	6797.08
	August	7278.11

Result Grid
Form Editor
Field Types

```

87 -- order by in product category with total revenue
88 • SELECT product_category, SUM(total_revenue) AS total_revenue
89 FROM sales_transactions
90 GROUP BY product_category
91 ORDER BY total_revenue desc;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

product_category	total_revenue
Electronics	34982.41
Home Appliances	18646.16
Sports	14326.52
Clothing	8128.93
Beauty Products	2621.90
Books	1861.93

```

93 -- only first 10 rows
94 • select * from sales_transactions limit 10;
95
96 -- top 5 product category revenue
97 • SELECT product_category, SUM(total_revenue) AS total_revenue
98 FROM sales_transactions
99 GROUP BY product_category

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

product_category	total_revenue
▶ Electronics	34982.41
Home Appliances	18646.16
Sports	14326.52
Clothing	8128.93
Beauty Products	2621.90